

Financial Transparency of Microfinance Institutions: Influence of Performance, Stakeholders, and Society

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Abstract

We investigate the determinants of microfinance institution (MFI) financial transparency, considering the impact of MFI performance and stakeholders, and societal characteristics. We find a positive association of transparency with MFI performance; however subsequent structural-equation modeling and other tests conclude that causality is in the direction of transparency levels determining portfolio quality. Regarding stakeholders, we find greater transparency is positively associated with MFIs being for-profit and negatively associated with taking deposits—even when controlling for prudential regulation and other factors. We conclude MFI transparency is positively responsive to for-profit shareholders but negative to depositor discipline. At the country-level we find MFI transparency positively associated with social trust and *de jure* regulation, and negatively associated *de facto* national governance, consistent with less consequences of unfavorable disclosure in environments of greater social trust and less national governance.

1. Introduction

International differences in microfinance continue to be of considerable interest (e.g., Bruton, Khavul and Chavez (2011) and Mersland, Randøy and Strøm (2011)). We examine the determinants of transparency of microfinance institutions (MFIs) around the world. Generally, with few exceptions (e.g., Nier and Baumann (2006)) the literature on disclosure focuses on non-financial corporations, with little attention to the banking sector. There is much less focus specifically on the microfinance sector. This study is of particular interest because transparency in the banking industry in general has been especially important to researchers and policy makers since the financial crisis of 2008 (e.g., Ng and Rusticus (2012)).

We categorize the determinants of MFI financial transparency in three ways: 1) performance, 2) stakeholders and 3) society. Each of these three categories are types of sounding boards for transparency. With regard to performance, firms can set disclosure levels in response to, for instance, profitability or efficiency or portfolio quality. While examinations of the association of performance and transparency, particularly with respect to the microfinance industry, have focused mainly on efficiency. In this paper we examine separately and jointly the impact of profitability, portfolio quality and efficiency on MFI transparency. While a large number of papers have in some fashion, either directly or indirectly, evidenced that transparency improves performance, few papers in finance have considered the possibility

of performance in turn influencing transparency. This is an especially noteworthy gap as literature from the social psychology literature (e.g., Greene, Derlega and Matthews (2006); Sprecher and Hendrick (2004)); as well as marketing literature (Crosby, Evans and Cowles (1990); Jacobs, Hyman and Mcquitty (2001)) is replete with examining the determinants of self-disclosure.

With regard to the second category, stakeholders, studying microfinance institutions affords a window into comparison of the behavior of for-profit versus non-profit firms. The behavior of non-profit firms as they compare to similar for-profit firms has been only lightly examined. For instance Adelino, Lewellen and Sundaram (2015) note that while nonprofit firms constitute a large portion of firms they have been comparatively only slightly researched compared to for-profit firms. Microfinance affords a unique opportunity to compare large numbers of for-profit and non-profit firms that are substantially similar in other respects. An examination of transparency at the MFI level affords an opportunity to examine how the determinants of MFI transparency, and banking transparency more generally, differ between profits and non-profits. We consider that for-profits will set volitional disclosure in response to shareholder concerns. Nonprofits will respond to donor concerns. Further, deposit-taking institutions will set disclosure in some measure in response to depositor discipline.

Lastly, while the importance of institutional and cultural factors on transparency has been examined to some extent (Gray (1988); Aggarwal and Goodell (2015)), there have been few if any studies examining the role of national governance and social trust on microfinance transparency in particular.¹ Generally, cross-national studies consistently show the importance of considering country-level factors, in addition to the firm-level characteristics. This is particularly true with regard to the importance of country-level factors in cross-national studies of corporate governance (Aslan and Kumar (2014); Filatotchev and Jackson (2013); and Kumar and Zattoni (2013)). Such studies illustrate that the relationship between corporate governance and firm performance is likely impacted by the quality of national governance. Further, researchers, particularly Barth, Caprio and Levine (2008b), Barth, Caprio

¹ An exception is Aggarwal, Goodell and Selleck (2015) who find for instance that MFIs target women borrowers more in environments of low social trust.

and Levine (2008a); Barth, Levine and Caprio (2008), have noted the important role of politics and institutional development in shaping both banking governance at the national level, and as a response, banking transparency at the bank level.

As noted recently by Bubna and Chowdhry (2010), issues of information sharing in the microfinance industry continue to be of critical importance. Transparency with regard to MFIs is of particular importance in developing countries with regard to the rapid changes in the microfinance industry. Cull, Demirgüç-Kunt and Morduch (2009) suggest that the microfinance industry as a whole will eventually evolve in some fashion from socially oriented nonprofit microfinance institutions to a more for-profit microfinance industry. But the path toward this will be multifaceted and the nature of microfinance will continue to be highly varied. Cull et al. (2009) suggest both for-profits and nonprofits are part of a vital mix of institutions seeking to facilitate financing to the bottom of the pyramid.

MFIs are similar to banks and traditional financial intermediaries. They intermediate funds between savers and borrowers. However, unlike traditional banking, MFIs generally do not depend in any significant way on consumer deposits. MFIs also differ from bank lending in other ways. For instance, microfinance involves small loans to the poor and relies on social networks and frequent repayments to reduce costs of monitoring and collection. Microfinance differs from traditional banking especially in the lack of collateral. Additionally, because of its unique social mission microfinance manifests differently from traditional banking in number of ways. For instance, portfolio at risk for microfinance institutions (MFIs) has traditionally been far lower than in the commercial banking sector. MFIs need stricter provisioning practices than banks or finance companies because their loans are less collateralized (von Stauffenberg, Jansson, Kenyon and Barluenga-Badiola (2003)). Banking laws usually do not take this into account and require provisioning policies and reserve levels that are inadequate for a microcredit portfolio.

Consequently, a study of the determinants of MFI transparency is important and can yield critical insights for policy-makers, and MFI institutions and consortiums. Further, despite considerable prior research on the role of national culture in determining accounting transparency and disclosure (e.g., Gray

(1988)), little previous research has applied such findings to investigate the determinants of MFI transparency and how it might be related to national culture and institutions, and MFI characteristics.

While there have been many studies of the macroeconomic and political impacts of the lack of the quality of MFI transparency, there seems to be no study of the determinants specifically of MFI transparency. MFI transparency is very important and there is sufficient reason to suspect that it is strongly affected by both MFI characteristics and national factors. This paper uses the transparency score of mixmarket.org. Using data on 1336 MFIs from 73 countries (6370 MFI/year observations), for the period 2003–2015, and controlling for relevant variables. We investigate the determinants of MFI financial transparency, considering the impact of MFI performance and stakeholders, and societal characteristics. We find a positive association of transparency with MFI performance; however subsequent structural-equation modeling and other tests conclude that causality is in the direction of transparency levels determining portfolio quality. Regarding stakeholders, we find greater transparency is positively associated with MFIs being for-profit and negatively associated with taking deposits—even when controlling for prudential regulation and other factors. We conclude MFI transparency is positively responsive to for-profit shareholders but negative to depositor discipline. At the country-level we find MFI transparency positively associated with social trust and *de jure* regulation, and negatively associated *de facto* national governance, consistent with less consequences of unfavorable disclosure in environments of greater social trust and less national governance.

Overall, we document results consistent with a picture that MFIs, working within the bounds of prudential regulations, expand disclosure to accommodate shareholders and in environments of low *de facto* national governance and high social trust. Further, MFIs respond to depositor discipline by lessening disclosure. MFIs subsequently set levels of portfolio quality to suitably match chosen disclosure levels.

2. Background

2.1 Importance of banking transparency

Transparency and information sharing has long been of great concern with regard to the banking industry in general—mainly because of the probability of bank runs (Morris and Shin (2002); Diamond and Dybvig (1983); Acharya, Gale and Yorulmazer (2011)). For instance, Angeletos, Hellwig and Pavan (2006) study the impact on banking outcomes of asymmetric information; Bryant (1980) finds that incomplete information has significant impact on the utility of demand deposits, reserves and deposit insurance. Research on disclosure and banking does not always conclude that more information is better. For instance, Siritto (2013) conclude that more precise information may increase economic instability by increasing incentives of depositors to withdraw funds, thereby increasing the probability of bank runs (see also Di Caprio and Santos-Arteaga (2012)). With regard to disclosure encouraging bank runs it comes down to whether transparency dominates the ability of markets and individuals to process information efficiency (Hirshleifer (1971); Dang, Gorton and Holmström (2009); Kiyotaki and Moore (1997); Kaplan (2006)). Although Amador and Weill (2010) present an alternative channel in which public information crowds out private signals. Bouvard, Chaigneau and De Motta (2015) alternatively consider whether bank disclosures may put some banks at greater risk for a run, but concomitantly lessen the probability of a systematic run on banks ensuing from public uncertainty.

These issues are part of a much broader research stream: whether ignorance dominates transparency in the provision of liquidity—with liquidity of course of particular concern for banking. Many papers suggest a seminal need for transparency in providing liquidity (Akerlof (1970)). Many studies (e.g., Goldstein, Hotchkiss and Sirri (2007); Edwards, Harris and Piwowar (2007), Bessembinder and Maxwell (2008) find that the cost of trading decreases with improved information flow.²

2.2 Microfinance transparency

Of course microfinance institutions differ from generally commercial banking in a number of distinctive and vital ways. While commercial banks may undertake microfinance as part of their broader

² Or alternatively when there is less cultural impetus for information (see Aggarwal and Goodell (2009)).

activities (and are increasingly doing so), microfinance is a specialized category of financial services offered to lower-income people, where the unit size of credit extension is “micro,” typically lower than the average GDP per capita (Isern and Porteous (2005)). Consequently, the discussion MFI disclosure will inevitably have a different focus from that of firm disclosure or banking disclosure in general.

Augustine (2012) notes that past research on the governance of MFIs (e.g., Mersland and Strøm (2009); Mersland and Strøm (2007)) has focused on the principal-agent problem. However, microfinance differs substantially from the simple shareholder view of ownership, with concomitant applicability of board and CEO theories to fully explain the corporate governance and transparency of MFIs. Further, while H notes that there has been little research on microfinance governance, even less has been on MFI transparency. H defines governance with regard to MFIs as the mechanisms through which donors, equity investors, and other providers of funds ensure themselves that their funds will be used according to the intended purposes. In this paper, we focus on the degree of transparency and disclosure in financial reporting. We define transparency, following Bushman, Piotroski and Smith (2004) as the availability of firm-specific information to those outside publicly traded firms. We extend this definition, in applying it to MFIs, to include both for-profit and non-profit firms. In other words, we do not restrict ourselves to publicly traded MFIs, but investigate the impact of MFI profit status while also examining other factors.

Bushman et al. (2004) makes the distinction between financial transparency, which captures the intensity and timeliness of financial disclosures, and governance transparency, which is the concomitant interpretation and dissemination of firm-level data by analysts, the media and the government; along with the necessary accountability of officers and directors. In this paper we investigate financial transparency as our dependent variable reflects MFI disclosure quality. We investigate whether this factor varies with MFI characteristics as well as national governance and other country-level factors.

2.3 Transparency, trust and governance

At least since Bushman et al. (2004), we have known that country-level factors influence corporate transparency. Bushman et al. (2004) find that more transparency is associated with better national governance. At least since Akerlof (1970), it has been recognized that asymmetric information

creates a cost to transactions. Grossman and Stiglitz (1980) affirm this, noting that the costs of information collection are a key component of transactions. Therefore, levels of disclosure have important implications for financial systems. According to North (1990), the costliness of information needed for measurement and enforcement of exchanges creates “transaction costs.” As noted by Williamson (1988) and many others (e.g., Hart (2001)), the primary transactions costs of contractual exchanges stem from asymmetric information and associated uncertainties.

Transaction Cost Economics suggests that the overall costs of exchanges have a significant impact on respective financial systems (Williamson (1975)). As noted by Bushman et al. (2004), high-quality³, provable information is integral to contracts.⁴ Further, idealistic assumptions aside, varying levels of governance (La Porta, Lopez-de-Silanes, Shleifer and Vishny (1998)) will impact the utility of reliable information. However, the nature of this impact deserves continuing study. For instance, transparency might increase with property rights because of a variety of reasons: 1) a concomitant reflection of governance; or 2) a greater demand for transparency when greater property rights make transparency more valuable (Aggarwal and Goodell (2015) and Milhaupt and Pistor (2008)). Of particular note here is the seminal study of Bushman et al. (2004) who find a positive association corporate transparency and the quality of national institutions.

Alternatively, transparency might be increased to compensate for lack of governance. Zer (2015) provides an interesting discussion of this possibility. Zer (2015) evidences that banks in particular increase their own disclosure when the public flow of information is poor. The results of Zer (2015), which are specific to banks, alongside those of Bushman et al. (2004) suggests the possibility that the relationship between transparency and national governance is different for banks than for firms in general.

³ Jacobs et al. (2001) note that self-disclosure can be either “exchange-specific” or “social” depending on whether or not such disclosure is relevant to the pending exchange

⁴ This point has been heavily emphasized in a number of articles examining the predilection for stock-market financing over bank financing. For example, Aggarwal and Goodell (2009), Kwok and Tadesse (2006), and Ergungor (2004) all suggest either explicitly or implicitly that the choice of stock market financing over bank financing will be influenced strongly by differences in transaction costs across nations, with these transaction costs primarily being costs associated with market contracts that are not present with relationship financing. Ergungor (2004) in particular finds markets are more favored over banks in environments of a common-law or English legal origin.

Generally, with few exceptions (e.g., Nier and Baumann (2006)) the literature on disclosure focuses on non-financial corporations, with little attention to the banking sector. There is much less focus specifically on the microfinance sector. An exception is Hartarska and Mersland (2012), although this paper is more on governance than specifically on disclosure. They document nuanced evidence that MFI efficiency increases with firm-level governance. They also find marginal evidence that MFIs regulated by an independent banking authority are more efficient, while MFIs in countries with “mature regulatory environments” reach fewer clients.

2.4 MFI Transparency and performance

In this section we discuss possible associations of transparency and MFI performance. We consider that transparency may impact performance; as well as consider whether performance impacts transparency. It is reasonable to consider that disclosure of bank balance-sheet details will incentivize banks to be more profitable, efficient or to hold less risky positions. This is because management wishes to maximize shareholder valuation. There is a presumption that withheld information is less favorable information (Grossman and Hart (1980); Grossman (1981); Lambert, Leuz and Verrecchia (2007, Milgrom (1981); Stiglitz (2002); Verrecchia (2001). Perhaps because of the impact of such additional incentives, many papers have documented an association of transparency and performance. For instance, Fonseca and González (2010) show better disclosures have positive effects on market discipline; lead to lower financing costs and lower risk profile. Other papers include Balakrishnan, Billings, Kelly and Ljungqvist (2014); Barth et al. (2013); Botosan and Plumlee (2002); Leuz and Verrecchia (2000) Bushee and Noe (2000); Diamond and Verrecchia (1991); Easley and O'Hara (2004); Kothari, Li and Short (2009). For the most part these papers evidence a strong positive association of firm-level costs of capital and investor valuations with improved disclosure and or more generally with improved governance.

2.5 Transparency, ownership and organization

Past research has noted differences between profit and non-profit firms, if not necessarily specifically focused on MFIs. For instance, Behn, DeVries and Lin (2010) note ongoing concerns in both the public (supervision) and private (donors) sectors regarding the transparency of nonprofits in the US.

Behn et al. (2010) examine the determinants of transparency in the nonprofit sector. They find that transparency for nonprofit is positively associated with being a larger organization, having more debt, a larger contribution ratio, being a higher-education nonprofit, and having a higher compensation expense ratio. In another example, Gordon, Fischer, Malone and Tower (2002); examine determinants of disclosure for higher education institutions in the US. They find an association public/private status were associated with levels of disclosure. Unfortunately, for drawing inferences regarding profit versus nonprofit as a determinant of transparency is clouded in this instance as private universities can be either for-profit or non-profit. More specific to microfinance, Beisland and Mersland (2014) examine determinants of differences in earnings quality between for-profit and nonprofit MFIs. However, they find insignificant differences.

As noted by Bhojraj, Blacconiere and Souza (2004), volitional transparency is a function of conflicting incentives amongst various stakeholders as audience. Additionally, in many industries there is differing financial accounting depending on for-profit versus non-profit status (Keating and Frumkin (2003)) as well as differing attitudes toward taxable ancillary activities (Yetman (2003)). Studying the disclosure of for-profit versus non-profit MFIs is an uncommon opportunity for further investigation of this distinction as a determinant of disclosure.

Duqi and Torluccio (2015) examine the impact of different ownership identities, such as shareholders, banks, social investors, government entities and institutional investors impact the risk and performance of MFIs. They find that different ownership forms are associated with conflicting goals with regard to the degree of emphasis placed on profitability versus, for instance, social performance.

Servin, Lensink and van den Berg (2012) investigate whether differences in MFI efficiency can be explained by differences in ownership. With a focus on non-governmental organizations, cooperatives and credit unions, non-bank financial intermediaries, and banks, they find that that non-governmental organizations and cooperatives have much lower interfirm and intrafirm technical efficiencies than non-bank financial intermediaries and banks, which indicates the importance of ownership type for technical efficiency. Mori and Mersland (2014) find that various stakeholders have differing influences on MFIs.

They find that differences in stake holders impact differences in the nature and focus of governing boards and also impact levels of financial and social performance.

2.6 Hypotheses

In this section we present several hypotheses regarding the expected association of transparency performance, national governance, social trust, and for-profit status. We consider that MFIs will be more willing to be transparent when they have better performance. In other words, transparency facilitates the advertising of success. As observed by Stiglitz (2002), transparency lessens the principal-agent problem caused by information asymmetries. Certainly from the point of view of the firm, there are less advantages to information asymmetry with investors when the firm is doing well (Akerlof (1970)). Additionally, with regard to individuals, Sprecher and Hendrick (2004) evidence, that there is greater self-disclosure within intimate relationships when individuals have greater self-esteem.

Alternatively, with regard to causality, but in keeping with the same association, transparency may improve performance. We cite a number of papers in a previous section noting the benefits of transparency to performance (e.g. Balakrishnan et al. (2014); Barth et al. (2013); Botosan and Plumlee (2002); Leuz and Verrechia (2000) Bushee and Noe (2000); Diamond and Verrechia (1991); Easley and O'Hara (2004); Kothari et al. (2009);)

H1: Transparency of MFIs is positively related to MFI performance

We consider that transparency is negatively related to national governance. We reason that when governance is less there will be correspondingly less consequences for unfavorable disclosure (Zer (2015)). Alternatively, transparency may be positively related to governance because of a general aptitude and trend for disclosure and governance throughout society (Bushman et al. (2004)).

H2: Transparency of MFIs is negatively related to national governance

H2a: Transparency of MFIs is positively related to national governance

According to the web page of the National Council of NonProfits, “as tax-exempt ‘public charities,’ charitable nonprofits embrace the values of accountability and transparency as a matter of ethical leadership, as well as legal compliance. Leaders of charitable nonprofits know that financial transparency will help preserve the very-important trust each donor places in a nonprofit with each contribution.”⁵ This would suggest an expectation of greater transparency amongst non-profits.

H3: Transparency of MFIs is positively related to for-profit status

However, it is also reasonable to expect that for-profit MFIs will disclose more in order to satisfy investors. We have discussed in a previous section the value of disclosure to investors. (e.g. Balakrishnan et al. (2014); Barth et al. (2013); Botosan and Plumlee (2002); Leuz and Verrechia (2000) Bushee and Noe (2000); Diamond and Verrechia (1991); Easley and O'Hara (2004); Kothari et al. (2009);)

H3a: Transparency of MFIs is negatively related to for-profit status

While it is widely considered that, across societies, transparency enhances trust, it is less clear that overall social trust will be impacted by microfinance transparency in particular. However, it may be possible that microfinance transparency is impacted by social trust. This is because it could be argued that it is safer to disclose information in environments of greater social trust. From the social-psychology literature, Sprecher and Hendrick (2004) evidence, for instance, that there is greater self-disclosure within intimate relationships when there is a greater overall quality of relationship.

H4: Transparency of MFI performance is positively related to social trust

3. Methodology

3.1 Statistical specification and estimation

Our initial empirical models and their estimates are based on the following Equation 1.

⁵ <https://www.councilofnonprofits.org/tools-resources/financial-transparency>, accessed May, 16, 2016

$$\begin{aligned} \text{TRANSPARENCY}_i = & \alpha + \sum \beta_1 \text{PERFORMANCE} + \sum \beta_1 \text{GOVERNANCE}_i \\ & + \beta_1 \text{TRUST} + \beta_2 \text{FOR_PROFIT} + \sum \beta_i \text{MFI_CONTROLS}_i \\ & + \sum \beta_i \text{MACRO_CONTROLS}_i + e_i \end{aligned} \quad 1)$$

In Equation 1, TRANSPARENCY_i is the transparency score for the MFI; PERFORMANCE_i is a vector of MFI-level variables specifically focused on the important business dimensions of efficiency, portfolio quality, and profitability. whether the MFI is for-profit or nonprofit, GOVERNANCE_i is a vector of national governance variables. These variables are described in the next sub-section in include aggregate for national governance from World Bank Indicators, a dummy for English legal origin and a variable that aggregates variables specific to banking regulation from Barth. TRUST is 100*(% most people can be trusted – % say you can't be too careful) from World Values Survey following Medrano (2011). With supplemental data from Medrano (2011) for some country/years. FOR_PROFIT is a dummy variable that is assigned “1” is the MFI is for-profit and “0” otherwise.

MFI_CONTROLS is a vector of is a vector of other MFI control variables such as the degree of outreach and size of assets, whether the bank accepts deposits, its MFI form whether NBO etc. other MFI-level controls relate to varying degree whether or not the MFI is subject to heightened prudential regulation.,

This multiple regression statistical model is estimated with standard errors clustered at the MFI level. Year dummies are included for the models as robustness checks and make no substantive difference to results. Variance inflation factors (VIF) are all less than ten for all variables for all models.

Our sample consists of 1336 microfinance institutions. The data is unbalanced panel data consisting of 6370 MFI/years from 73 countries. Our time period is 2003–2015.

3.2 Dependent variable

We use data from www.mixmarket.org assessment of MFI transparency (TRANSPARENCY). The mix transparency score (labeled as “diamonds” in their dataset) is assigned 0,1,2,3,4, or 5, with 5 being most transparent. Currently, each profile on the site receives an overall profile diamond score and an annual score for each full year of data that is published on the site. A diamond score of “1” is assigned

if the MFI is considered to have a visible profile. A diamond score of “2” is assigned if MFI reports some data on products and clients for the year. A diamond score of “3” is assigned if there is additionally some reporting of financial data. A diamond of “4” is assigned if additionally audited financial statements are published for the year. A diamond of “5” is awarded if MFI additionally publishes a due diligence report for the year.

Table 1 shows the country averages in our sample, ranked according to transparency. Also shown is the number of MFIs and MFI/years per country. Also shown are relative numbers of profit versus non-profits MFIs and MFI/years; as well relative numbers of MFIs and MFI/years for deposit-taking versus non-deposit-taking institutions.

Examining Table 1, we find that transparency varies widely across our broad sample of MFIs of 73 countries. The country with the best transparency, perhaps surprisingly, is Paraguay (5.00). Panama and Cambodia also rank relatively high on transparency. India has the most MFIs per country (539 out of 6370). Transparency of MFIs from Eastern Europe and Africa seem to be relatively lower.

(Please insert Table 1 about here)

3.3 Independent variables

The independent variables can be classified into several categories: MFI performance characteristics, MFI control variables, governance, trust, and macroeconomic variables. Next we describe the composition of each of these categories.

3.3.1 MFI performance characteristics

Following, von Stauffenberg et al. (2003), regarding MFI financial characteristics we assess determinants of transparency based on four fundamental MFI qualities 1) portfolio quality or portfolio risk, 2) efficiency and productivity, 3) financial management and 4) profitability.

For portfolio quality (PORTFOLIO_QLTY) we use the principal component of Principal component of portfolio at risk 90-days, write off ratio, $-1 \times$ provision expense ratio, $-1 \times$ risk coverage from mixmarket.org. Provision expense ratio is provision for loan impairment/ gross loan portfolio. Risk coverage is impairment loss allowance/ 30-day portfolio at risk.

For efficiency and productivity (EFFICIENCY) we use the principal component of the operating expense ratio, cost per borrower, personnel productivity and loan-officer productivity. For financial management we would prefer to use the principal component of the funding expense ratio, cost of funds ratio and the debt-to-equity ratio. However, mixmarket.org does not maintain data for the funding expense ratio and the cost of funds ratio. So we use only the debt-to-equity ratio. We label this variable “LEVERAGE.” For profitability (PROFITABILITY) we use the principal component of return on equity, return on assets, and portfolio yield. We also test an overall measure of performance (PERFORMANCE) as the first principal component of PORTFOLIO_QUALITY, EFFICIENCY, and PROFITABILITY.

3.3.2 MFI control variables

Additionally, we control for several MFI mission characteristics. We control for MFI size with the natural log of the size of assets (LNASSETS). This is reported by mixmarket.org in US dollars. We also control for MFI age (AGE). We also include as an independent variable a measure degree of outreach from mixmarket.org (OUTREACH). This variable is assigned 0,1, or 2, with 2 being the highest level of outreach, with outreach proxied by the number of borrowers with “2” assigned if this number is greater than 30,000; “1” assigned if this number is between 10,000 and 30,000 and “0” if less than 10,000.

We also control for the legal status (REGULATED) and type (CHARTER_TYPE) of MFIs. REGULATED is a dummy variable that is assigned “1” if the MFI as an entity is submitted to some regulatory authority, whether a formal banking regulator or some other financial services regulator. This most often concerns entities that are listed as bank’ or non-bank financial institutions (NBFIs), but may also include credit union/cooperatives’ or non-governmental organizations (NGOs). REGULATED is assigned “0” otherwise. CHARTER_TYPE is a dummy variable that is assigned a “1” if the MFI is a bank, credit union, cooperative bank or rural bank and is assigned a “0” if MFI is a NBFI, NGO or undefined. SUSTAINABILITY is a dummy A dummy variable the is assigned “1” if MFI has 100% financial self-sufficiency and “0” if financial self-sufficiency less than 100%. DEPOSITS is a dummy variable that is assigned “1” if the MFI accepts deposits and “0” otherwise. TARGET is a categorical variable that is assigned an integer 1–4 depending on whether MFI is 1) “low end,” with depth less than

20% or average loan size is less than \$150; 2) “broad,” with depth between 20% and 149%; 3) “high end,” with depth between 150% and 250%; or “small business,” with depth over 250%.

3.3.3 Governance and trust variables

In order to assess the impact of culture on MFI transparency, we include TRUST This variable is formed as $100 * (\% \text{ most people can be trusted} - \% \text{ say you can't be too careful})$ from World Values Survey following Medrano (2011). With supplemental data from Medrano (2011) for some country/years.

BANKING_REGULATION is the principal component of six dummy variables: 1) Are banks required to prepare consolidated accounts for accounting purposes? 2) Are applicable accounting standards for banks in your country prepared in accordance with U.S. Generally Accepted Accounting Principles (GAAP) at individual bank level? 3) Are applicable accounting standards for banks in your country prepared in accordance with IFRS at individual bank level? 4) Are all banks operating in your country (including foreign bank branches) required to make available to the public their annual financial statements at individual bank level? 5) Do banks disclose to the supervisors fully audited financial statements? 6) Are bank directors legally liable if information disclosed is erroneous or misleading? These variables are from World Bank’s Banking Regulation and Supervision Database of 2012. Finally, we also control for the national macroeconomic environment with the natural log of GNI per capita (LN_GNI_CAP) and the annual percentage rate of growth in GNI per capita (GNI_CAP_GROWTH).

3.3.4 Macroeconomic variables

Table 2 is a summary of the dependent and independent variables used in this study. Together these independent variables reflect the influences that may affect MFI predilection for transparency. As noted earlier, we independent variables that reflect fund financial characteristics, fund mission characteristics, national culture, national institutions, and macroeconomic factors.

(Please insert Table 2 about here)

4. Results

4.1 Determinants of MFI transparency: The impact of MFI performance

In our first results table, Table 3, we report regression models that focus on the determinants of transparency. Our set of independent variables in Models 1–4 includes the variable of particular interest PERFORMANCE, which is the first principal component of PROFITABILITY, EFFICIENCY and PORTFOLIO_QLTY. We include a wide of MFI control variables in each model. In Models 2–4 we add in turn macroeconomic variables, national governance variables and trust. All models have variance inflation factors (VIF) of less than 10 for all regressors indicating that multicollinearity is unlikely to be a problem. Because of the nature of our dependent variable which can take on the values of 0,1,2,3,4 or 5, we report results of ordered logit regressions with errors clustered at the MFI level and controlling for year fixed effects.

(Please insert Table 3 about here)

Overall, our results for the determinants of transparency for our combined sample of 1336 MFIs from 73 countries (6370 MFI/year observations), is that MFIs are more transparent when they have better performance. PERFORMANCE is positively significant in each model—although PERFORMANCE is significant only at 10% in the most comprehensive Model 4. These results are consistent with MFIs choosing to be more transparent when they have better performance. In other words, MFIs are more willing to disclose their details when they have more to boast about. Alternatively, as these empirical tests investigate association rather than causation, the positive significance of performance could be due to greater transparency causing better performance (Akhigbe, McNulty and Stevenson (2013); Nier and Baumann (2006); Diamond and Verrechia (1991)). We investigate this question further in a subsequent section that presents structural-equation modeling.

Regarding macroeconomic factors, we find transparency to be negatively related to the natural log of GNI per capital and to the growth rate of GNI per capita in Model 2. This suggests that in economies performing better, where microfinance is likely less needed, MFIs are less transparent. However, in Models 3 and 4, while GNI_CAP_GROWTH continues to be negatively significant, LN_GNI_CAP is not significant.

Regarding national governance variables, our results suggest the transparency of MFIs responds very differently with regard to *de facto* governance measures as opposed to *de jure* measures. MFI transparency is positively significantly associated with BANK_REGULATION which is formed from various *de jure* measures of banking disclosure requirements as described above and in Table 2. On the other hand, MFI transparency is significant negatively associated with our measure of *de facto* governance from the World Bank. World Bank governance indicators are survey generated and assess impressions of governance rather than laws on the books. Overall we interpret our results for governance as suggesting MFIs will disclose more information when specifically directed by regulations. They will also seek greater transparency in environments of less governance quality. This latter result is consistent with MFIs possibly seeking to compensate to their stakeholders for a lack of national governance; or, alternatively, MFIs willing to disclosure more in polity environments in which there are less consequences for negative disclosure. Our concomitant result in Model 4 of social trust being positively significant is consistent with this latter interpretation. Greater trust implies less risks of disclosure.

4.2 Determinants of MFI transparency: Examining the components of performance

In this section we re-do our testing of the comprehensive Model 4 of Table 3, but alternatively decompose our measure of performance into the components of profitability, portfolio quality and efficiency. We test these three measures separately and then all together in the models of Table 4. The models of Table 4 clearly show that the driver of our results for PERFORMANCE of Table 3 is portfolio quality. PORTFOLIO_QLTY is positively significant, both standing alone as a performance measure and when PROFITABILITY and EFFICIENCY are included in the set of independent variables. MFIs are more willing to disclosure information about themselves when they have better portfolio quality.

(Please insert Table 4 about here)

4.3 Structural-equation analysis

The results of the previous section regarding the positive association of transparency and portfolio quality leave open the question of the direction of causality. Are MFIs more will to undertake disclosures when they have better portfolio quality? Or, alternatively, does transparency itself positively impact portfolio quality? In order to investigate this question, we model the following set of structural-equations (Equation 2).

$$\begin{aligned} \text{TRANSPARENCY}_i &= \alpha_i + \beta * \text{PORTFOLIO_QLTY} + \sum \beta_i * \text{MFI_PERFORMANCE}_i + \sum \beta_i * \text{MFI_CONTROLS}_i \\ &+ \sum \beta_i * \text{GOVERNANCE}_i + \sum \beta_i * \text{MACROECONOMIC}_i + e_i \end{aligned} \quad 2)$$
$$\begin{aligned} \text{PORTFOLIO_QLTY} &= \alpha_i + \beta * \text{TRANSPARENCY} + \sum \beta_i * \text{MFI_PERFORMANCE}_i \\ &+ \sum \beta_i * \text{MFI_CONTROLS}_i + \sum \beta * \text{MACROECONOMIC} + e_i \end{aligned}$$
$$\text{TRUST} = \alpha_i + \beta * \text{WORLD_BANK_GOVERNANCE} + \beta * \text{ENGLISH_LEGAL} + e_i$$

Results for this modeling are reported in Table 5. Of particular interest are the results for the relationship between transparency and portfolio quality. Unlike the models of Tables 3 and 4, PORTFOLIO_QLTY is not a significant determinant of TRANSPARENCY. However, TRANSPARENCY is a significant determinant of PORTFOLIO_QLTY. We interpret this result as consistent with greater transparency improves portfolio quality. In other words, if you know you must disclose your portfolio information, you will want to have good portfolio quality.⁶

Results for the structural-equation modeling suggest MFI control variables have their own direct roles in determining both transparency and portfolio quality—as opposed to indirectly impacting transparency through portfolio quality or indirectly impacting portfolio quality through transparency. For example, several MFI control variables have a negative impact on portfolio quality while having a

⁶ As noted by Bollen and Pearl (2013), inference of causality from structural-equation analysis is controversial. However, Bollen and Pearl (2013) posits that when concomitantly applied with reasonable theoretical foundation, structural-equation modeling strongly adds to our understanding of likely causal relationships. In this study we have looked to structural-equation modeling for insights into causal relationships rather than Granger causality tests because of low time variation of our measure of transparency. While transparency is MFI/year data, there are relatively few changes in this variable across the years of our period of study for particular MFIs. And so Granger Causality testing is difficult to employ.

positive impact on transparency (FOR_PROFIT and LNASSETS) or have a positive impact on portfolio quality and a negative impact on transparency (CHARTER_TYPE). This is inconsistent with indirectly impacting transparency by first impacting portfolio quality or *vice versa*. Otherwise, we would expect those variables with a negative association with portfolio quality to also have a negative association with transparency etc.

As an example, our results are consistent with for-profit status lowering portfolio quality but also increasing transparency. This suggests for-profit MFIs in an effort to increase profits extend credit to the lowering of portfolio quality, while at the same time have less concerns regarding disclosure. While not reported in the tables, we also test the impact of the interaction of for-profit status and portfolio quality on transparency, finding this variable to be negatively significant. This result is consistent with for-profit status lessening the impact of portfolio quality on transparency. On the other hand, the interaction of for-profit status and transparency is insignificant in determining portfolio quality. This is consistent with portfolio quality of for-profit MFIs being driven by their business model and not in reaction to levels of disclosure. However, this result is possibly inconsistent with Nier and Baumann (2006) who suggest that bank disclosure leads to market discipline which in turn contains bank risk taking. Under the argument of Nier and Baumann (2006), one might expect for-profits to have greater sensitivity of transparency to portfolio quality as they are subject to market discipline. Results suggest non-profits are subject to other forms of stakeholder discipline. On the other hand, Hartarska (2005) find that having a donor on the board does not improve the efficiency of MFIs.

(Please insert Table 5 about here)

4.4 Robustness testing with propensity score matching

In this section we control with propensity matching whether our results for DEPOSITS and FOR_PROFIT are influenced by profitability, leverage and size influencing whether an MFI takes deposits or influences choice to be FOR_PROFIT. We use propensity score matching to control for this possible endogeneity. In Model 1 of Table 6 we use MFIs that accept deposits as a treatment group and

MFIs that do not accept deposits as a control group. Similarly, in Model 2 we use MFIs that are for-profit as a treatment group and MFIs that are nonprofit as a control group. We match each of these MFI pairs of treatment and control based on profitability, size and leverage using a non-replacement approach.

Results of Table 6 confirm our earlier results. There are no substantive changes to the signs and significances of the independent variables from the models of other tables—even though propensity-score matching naturally reduces the sample size considerably. Portfolio quality, MFI size, for-profit status, banking regulation and social trust are positively significant. MFI age, sustainability, charter status, deposit taking, and national governance are negatively significant.

(Please insert Table 6 about here)

4.5 Discussion

The primary finding of this paper with regard to performance as a determinant of transparency is a positive association of microfinance institution transparency and particularly respective portfolio quality. Our finding of a positive association of transparency and performance is driven by an association of portfolio quality and not with efficiency or profitability. However, with structural-equation modeling, based on plausible causal assumption (Bollen and Pearl (2013)), With other controls we simultaneously model portfolio quality as determining transparency as well as transparency determining portfolio quality. This results in portfolio quality not being significant in determining transparency, but on the other hand transparency is significant in determining portfolio quality. We interpret this result as due to transparency compelling portfolio quality, rather than MFIs choosing greater transparency in response to better portfolio quality. Using our taxonomy, this points to stakeholder and country factors setting transparency with portfolio quality then set to meet this level of transparency.

With regard to MFI stakeholders, a notable result of this study is that we find whether an MFI accepts deposits (DEPOSITS) is negatively significant in every model in which it is present. Protecting the safety of deposits provides a rationale for improved regulation and supervision of microfinance institutions. Consequently, Christen, Lauer, Lyman and Rosenberg (2012) posit that prudential

regulations is usually triggered when an MFI accepts retail deposits from the general public (see also Cull, Demircuc-Kunt and Morduch (2011)). Therefore, results for DEPOSITS might be considered in keeping with MFIs being less transparent when faced with prudential regulation. However, our variable for prudential regulation REGULATED is not significant. We also control for charter status, finding that CHARTER_TYPE is negatively significant in every model in which it is present.

With regard to our hypotheses, we do find support for *H1*, that MFI transparency is positively related to performance. However, based on structural-equation analysis and other testing, we determine that this positive relation is likely to a direction of causality of transparency determining performance, with the impact of transparency mainly due to a setting of portfolio quality. With regard to *H2*, we do find support that MFI transparency is negatively related to national governance. With regard to *H3*, we find support for a positive association of MFI transparency and for-profit status. With regard to *H4*, we do find a positive relation between MFI transparency and social trust.

Overall, results suggest that the negative significance of DEPOSITS is due to another reason than implying prudential regulation or a particular charter. Deposit-taking MFIs are less transparent—even when controlling for these other factors.⁷

In contrast to the reaction of transparency to the accepting of deposits, our dummy variable for whether an MFI is for-profit is positively significant. This result is consistent with MFI transparency being improved by private shareholder monitoring but not by depositor discipline. With regard to country-level factors, we find MFI transparency positively associated with social trust and *de jure* regulation, and negatively associated *de facto* national governance, consistent with less consequences of unfavorable disclosure in environments of greater social trust and less national governance.

Regarding other MFI results, TARGET is also negatively significant in determining both transparency and portfolio quality. This suggests MFIs with greater loan depth are less transparent, possibly because of less portfolio quality. SUSTAINABILITY is mildly negatively significant in

⁷ Correlation between DEPOSITS and REGULATED is 0.35; between DEPOSITS and CHARTER_TYPE is 0.49, as reported in Appendix 1

determining transparency in Tables 3 and 4. However, in Table 5, sustainability negatively determines portfolio quality (at 1%) but is not significant in determining transparency. GNI_CAP_GROWTH is negatively significant in a number of models. A negative association of MFI transparency and economic growth could be due to the rather common story of economic expansion leading a shift toward less portfolio quality.

5. Conclusions

International differences in microfinance continue to be of considerable interest (e.g., Bruton et al. (2011); Mersland et al. (2011)). We examine the cross-national determinants of the transparency of microfinance institutions. Transparency in the banking industry in general has been especially important to researchers and policy makers, especially since the financial crisis of 2008. Further, transparency with regard to MFIs is of particular importance considering the rapid changes in the microfinance industry. Cull et al. (2009) suggest that the microfinance industry as a whole will eventually evolve in some fashion from socially oriented nonprofit microfinance institutions to a more for-profit microfinance industry. But the path toward this will be multifaceted and the nature of microfinance will continue to be highly varied. Consequently, this study should be of great interest to policy makers and scholars interested especially in the design and operation of microfinance institutions.

At the country-level we find MFI transparency positively associated with social trust and *de jure* regulation, and negatively associated *de facto* national governance. We interpret these results as due to less consequences of unfavorable disclosure in environments of greater social trust and less governance. At the firm level we first find a positive association of transparency with MFI performance. However, additional investigation with suggest this association of performance and transparency is driven by portfolio quality. We further investigate with structural-equation analysis whether a positive association of transparency and portfolio quality is due to MFIs willing to disclose more when they are doing better, or alternatively whether more transparency leads to better MFI performance. We document results consistent with the latter explanation, that transparency leads to better portfolio quality.

We also find that MFI financial transparency is greater with respect to for-profit MFIs than for nonprofit MFIs and that disclosure is less for MFIs taking deposits even when controlling for prudential regulation other factors. Overall, we conclude that MFIs, working within the bounds of prudential regulations, expand disclosure to accommodate shareholders and in environments of low de facto national governance and high social trust. Further, MFIs respond to depositor discipline by lessening disclosure. MFIs then set levels of portfolio quality to suitably match chosen disclosure levels.

These results should be of great interest to researchers and policy makers interested in banking transparency, and particularly in the transparency of the microfinance industry as the mix of the industry evolves to a mix of for-profit and nonprofit institutions.

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Table 1: Transparency of MFIs by country

This table reports country-level averages of transparency of MFIs. Transparency is assigned 0,1,2,3,4,5 with 5 being most transparent from mixmarket.org

	Country	All MFIs			For Profit MFIs		Non-Profit MFIs	
		Transparency	MFI-yrs	MFIs	Transparency	MFI-yrs	Transparency	MFI-yrs
1	Paraguay	5.00	45	6	5.00	31	5.00	14
2	Panama	4.29	34	4	4.67	15	4.00	19
3	Cambodia	4.13	132	18	4.13	130	4.00	2
4	Bolivia	4.04	209	25	4.18	89	3.94	120
5	Ecuador	4.03	413	58	4.53	86	3.90	327
6	Mongolia	3.97	38	7	3.97	33	4.00	5
7	Bosnia Herzegovina	3.83	126	14	3.77	26	3.85	100
8	Congo	3.83	24	7	3.50	2	3.86	22
9	India	3.79	539	134	4.20	360	2.98	179
10	Azerbaijan	3.79	131	22	3.85	61	3.74	70
11	Honduras	3.79	162	25	3.81	67	3.78	95
12	Georgia	3.76	83	15	3.96	45	3.53	38
13	Serbia	3.72	32	4	3.61	23	4.00	9
14	Papua New Guinea	3.67	12	2	3.67	12	*	*
15	Tajikistan	3.66	116	27	3.79	75	0.00	41
16	El Salvador	3.64	121	15	3.74	61	3.55	60
17	Nicaragua	3.60	213	29	4.18	93	3.15	120
18	Kazakhstan	3.58	60	13	3.69	51	3.00	9
19	Guatemala	3.54	154	22	4.00	1	3.54	153
20	Kosovo	3.52	62	10	2.91	11	3.65	51
21	Colombia	3.51	193	36	4.29	49	3.25	144
22	Nepal	3.51	125	30	3.38	78	3.72	47
23	Argentina	3.51	61	16	4.14	35	2.65	26
24	South Africa	3.50	10	5	3.63	8	3.00	2
25	Ukraine	3.50	12	2	1.00	2	4.00	10
26	Armenia	3.47	74	11	3.58	55	3.16	19
27	Afghanistan	3.47	53	15	4.00	8	3.38	45
28	Kenya	3.46	65	16	3.47	45	3.45	20
29	Nigeria	3.45	40	15	3.44	32	3.50	8
30	Chile	3.39	28	7	4.07	15	2.62	13
31	Mexico	3.37	315	69	3.51	269	2.57	46
32	Macedonia	3.36	28	4	1.00	3	3.64	25
33	Pakistan	3.35	159	28	3.40	87	3.28	72
34	China	3.26	27	6	3.00	12	3.47	15
35	Bangladesh	3.25	182	35	4.88	24	3.01	158
36	Kyrgyzstan	3.24	88	20	3.45	77	1.73	11
37	Dominican Republic	3.23	56	10	2.54	24	3.75	32
38	Benin	3.19	52	15	1.00	2	3.28	50
39	Haiti	3.12	43	8	4.00	22	2.19	21
40	Brazil	3.05	128	35	2.89	27	3.09	101

Table 1: Transparency of MFIs by country

This table reports country-level averages of transparency of MFIs. Transparency is assigned 0,1,2,3,4,5 with 5 being most transparent from mixmarket.org

	Country	All MFIs			For Profit MFIs		Non-Profit MFIs	
		Transparency	MFI-yrs	MFIs	Transparency	MFI-yrs	Transparency	MFI-yrs
41	Burundi	3.00	6	3	3.00	2	3.00	4
42	Peru	2.99	351	63	3.41	144	2.70	207
43	Romania	2.97	37	7	2.75	28	3.67	9
44	Jordan	2.96	55	8	3.00	6	2.96	49
45	Lebanon	2.95	21	4	2.89	9	3.00	12
46	Senegal	2.86	43	11	4.56	9	2.41	34
47	Cote d'Ivoire	2.82	11	8	2.67	6	3.00	5
48	Albania	2.78	36	6	2.75	32	3.00	4
49	Sudan	2.75	4	4	3.33	3	1.00	1
50	Rwanda	2.69	32	7	2.71	21	2.64	11
51	Sri Lanka	2.65	62	20	3.00	42	1.90	20
52	Malawi	2.61	23	5	3.00	5	2.50	18
53	Liberia	2.50	4	2	3.00	3	1.00	1
54	Indonesia	2.43	138	44	3.00	72	1.82	66
55	Mozambique	2.40	42	8	2.40	15	2.41	27
56	Uzbekistan	2.38	26	13	3.00	18	1.00	8
57	Zambia	2.35	17	6	1.73	11	3.50	6
58	Bulgaria	2.28	88	22	2.64	11	2.23	77
59	Ethiopia	2.25	76	16	2.22	59	2.35	17
60	Tanzania	2.25	44	11	2.00	13	2.35	31
61	Philippines	2.11	340	76	1.73	162	2.46	178
62	Niger	2.05	19	6	1.00	4	2.33	15
63	Madagascar	2.05	39	12	2.43	14	1.84	25
64	Russia	2.05	165	56	2.45	29	1.96	136
65	Burkina Faso	2.04	25	6	3.00	11	1.29	14
66	Poland	1.92	13	3	2.33	9	1.00	4
67	Sierra Leone	1.91	11	6	2.20	5	1.67	6
68	Ghana	1.84	89	33	1.42	57	2.59	32
69	Uganda	1.83	53	12	1.49	41	3.00	12
70	Cameroon	1.81	26	9	1.47	19	2.71	7
71	Guinea	1.73	11	3	3.00	4	1.00	7
72	Yemen	1.15	13	5	2.00	2	1.00	11
73	Thailand	1.00	5	1	1.00	5	*	*
Total		3.27	6370	1336	3.46	3017	3.11	3353

Table 2: Descriptive Statistics and Summary of Data Sources

This table lists the mean, standard deviations, minimum and maximum; and sources of variables used in regressions which are reported in Tables 3, 4 and 5. Unbalanced panel data for 2003–2015. 1336 MFIs from 73 countries (6370 MFI/year observations).

Variable	Obs	Mean	Stdev	Min	Max	Source
TRANSPARENCY	6391	3.27	1.35	0	5	Transparency. Assigned 0,1,2,3,4,5 (mixmarket.org). 5 is most transparent
EFFICIENCY	6391	−0.00	1.15	−45.53	19.04	−1*(Principal component of operating expense to loan portfolio, cost per borrower, borrowers per staff member, borrowers per loan officer) From mixmarket.org
PORTFOLIO_QLTY	6391	0.00	1.10	−51.23	6.04	−1*(Principal component of portfolio at risk 90days, write off ratio, −1*provision expense ratio, −1*risk coverage). From mixmarket.org. Provision expense ratio is provision for loan impairment/ gross loan portfolio. Risk coverage is impairment loss allowance/ 30-day portfolio at risk.
PROFITABILITY	6391	−0.00	1.07	−38.28	3.96	Principal component of return on equity, return on assets, yield on gross portfolio nominal from mixmarket.org
PERFORMANCE	6391	0.00	1.17	−37.86	22.33	Principal component of standardized EFFICIENCY, PORTFOLIO_QLTY and PROFITABILITY
LEVERAGE	6391	6.46	37.98	0.00	1436.6	Ratio of debt to equity (mixmarket.org)
LNASSETS	6392	16.14	1.80	10.92	23.99	Natural log of MFI assets (mixmarket.org)
FOR_PROFIT	6391	0.47	0.50	0	1	A dummy variable that is assigned “1” if MFI is for profit and “0” otherwise (mixmarket.org)
OUTREACH	6391	0.88	0.86	0	2	The degree of outreach. Assigned 0,1,2 (mixmarket.org). 2 is most outreach. (mixmarket.org)
SUSTAINABILITY	6391	0.78	0.41	0.00	1.00	A dummy variable the is assigned “1” if MFI has 100% financial self-sufficiency and “0” if financial self-sufficiency less than 100%. (mixmarket.org)
REGULATED	6373	0.66	0.48	0.00	1.00	Yes=1, No=0 This field is marked as ‘Yes’ if the entity is submitted to some regulatory authority, whether a formal banking regulator or some other financial services regulator. This most often concerns entities that are listed as ‘Banks’ and ‘Non-Bank Financial Institutions (NBFIs)’, but may also include ‘Credit Union / Cooperatives’ or ‘Non-Governmental Organizations (NGOs)’ in some markets. (mixmarket.org)
DEPOSITS	6391	0.47	0.50	0.00	1.00	Accepts deposits = 1, No deposits = 0 (mixmarket.org)
CHARTER_TYPE	6388	0.28	0.45	.00	1.00	Bank, Credit Union / Cooperative, Rural Bank=1, NBFi, NGO, Other=0 (mixmarket.org)
TARGET	6391	0.55	0.50	0.00	1.00	High-end & Broad=1, Low-end & Small business=0 (mixmarket.org)
LN_GNI_CAP	6391	10.91	2.30	5.58	17.51	Natural log of GNI per capita from World Development Indicators
GNI_CAP_GROWTH	6391	4.37	5.78	−15.51	63.41	Annual growth in GNI per capita from World Development Indicators
TRUST	5700	0.43	17.21			100*(% most people can be trusted – % say you can’t be too careful) from World Values Survey following Medrano (2011). With supplemental data from Medrano (2011) for some country/years.
WORLD_BANK_GOVERNANCE	6367	0.00	1.95	−6.36	9.53	Principal component of six governance factors of World Bank
ENGLISH_LEGAL	6391	.24	.43	0.00	1.00	A dummy variable that is assigned “1” is country has an English legal origin and “0” otherwise.
BANK_REGULATION	5503	0.00	1.22	−4.28	2.11	Principal component of 6 dummy variables: 1) Are banks required to prepare consolidated accounts for accounting purposes? 2) Are applicable accounting standards for banks in your country prepared in accordance with U.S. Generally Accepted Accounting Principles (GAAP) at individual bank level? 3) Are applicable accounting standards for banks in your country prepared in accordance with IFRS at individual bank level? 4) Are all banks operating in your country (including foreign bank branches) required to make available to the public their annual financial statements at individual bank level? 5) Do banks disclose to the supervisors fully audited financial statements? 6) Are bank directors legally liable if information disclosed is erroneous or misleading? From World

Table 3: Determinants of MFI transparency:

Table shows results of ordered logit regressions with standard errors clustered by microfinance institution, controlling for year fixed effects. Dependent variable, TRANSPARENCY, can take on discreet values of 0, 1, 2, 3, 4, 5. Variance inflation factors for all variables in all models less than 10. Variables defined in Table 2. P values in parentheses. Unbalanced panel data for 2003–2015. 1336 MFIs from 73 countries (6370 MFI/year observations).

	TRANSPARENCY			
	1	2	3	4
PERFORMANCE	0.16*** (0.008)	0.17*** (0.009)	0.17** (0.018)	0.13* (0.073)
LEVERAGE	−0.00 (0.253)	−0.00 (0.277)	−0.00 (0.451)	−0.00 (0.368)
LNASSETS	0.28*** (0.000)	0.27*** (0.000)	0.32*** (0.000)	0.33*** (0.000)
AGE	−0.16** (0.042)	−0.18** (0.028)	−0.23** (0.009)	−0.23** (0.020)
FOR_PROFIT	0.32** (0.011)	0.36*** (0.005)	0.35** (0.016)	0.28* (0.078)
OUTREACH	0.07 (0.416)	0.10 (0.286)	0.10 (0.346)	0.11 (0.355)
SUSTAINABILITY	−0.21* (0.059)	−0.21* (0.056)	−0.25** (0.036)	−0.25** (0.048)
REGULATED	−0.07 (0.603)	−0.07 (0.590)	0.03 (0.835)	0.05 (0.756)
DEPOSITS	−0.61*** (0.000)	−0.56*** (0.000)	−0.67*** (0.000)	−0.51*** (0.001)
CHARTER_TYPE	−0.46*** (0.003)	−0.45*** (0.005)	−0.59** (0.001)	−0.74*** (0.000)
TARGET	0.07 (0.525)	0.05 (0.657)	−0.12 (0.337)	−0.08 (0.584)
LN_GNI_CAP		−0.06** (0.051)	−0.05 (0.147)	0.02 (0.614)
GNI_CAP_GROWTH		−0.02*** (0.001)	−0.02 (0.020)	−0.01* (0.071)
WORLD_BANK_GOVERNANCE			−0.20*** (0.000)	−0.14*** (0.006)
ENGLISH_LEGAL			−0.29* (0.070)	−0.22 (0.185)
BANK_REGULATION			0.33*** (0.000)	0.41*** (0.000)
TRUST				0.01*** (0.008)
YEAR DUMMIES		YES		
Observations	6370	6370	5459	4993
Wald	682.79*** (0.000)	708.09*** (0.000)	647.34*** (0.000)	620.32*** (0.000)

*significant at 10% level, **significant at 5% level; ***significant at 1% level

Table 4: Determinants of MFI transparency: Impact of components of performance

Table shows results of ordered logit regressions with standard errors clustered by microfinance institution, controlling for year fixed effects. Dependent variable, TRANSPARENCY, can take on discrete values of 0,1,2,3,4,5. Variance inflation factors for all variables in all models less than 10. Variables defined in Table 2. P values in parentheses. Unbalanced panel data for 2003–2015. 1336 MFIs from 73 countries (6370 MFI/year observations)

	TRANSPARENCY			
	1	2	3	4
PROFITABILITY	0.05 (0.620)			0.00 (0.989)
PORTFOLIO_QLTY		0.16*** (0.000)		0.16*** (0.000)
EFFICIENCY			−0.07 (0.423)	−0.07 (0.448)
LEVERAGE	−0.00 (0.377)	−0.00 (0.336)	−0.00 (0.363)	−0.00 (0.331)
LNASSETS	0.33*** (0.000)	0.33*** (0.000)	0.34*** (0.000)	0.34*** (0.000)
AGE	−0.22** (0.029)	−0.20** (0.034)	−0.22** (0.027)	−0.21** (0.033)
FOR_PROFIT	0.25 (0.107)	0.26* (0.097)	0.26 (0.101)	0.27* (0.086)
OUTREACH	0.12 (0.292)	0.12 (0.319)	0.10 (0.407)	0.10 (0.425)
SUSTAINABILITY	−0.17 (0.196)	−0.20* (0.082)	−0.16 (0.188)	−0.22* (0.074)
REGULATED	0.07 (0.677)	0.07 (0.681)	0.07 (0.681)	0.06 (0.693)
DEPOSITS	−0.53*** (0.000)	−0.54*** (0.000)	−0.53*** (0.000)	−0.53*** (0.000)
CHARTER_TYPE	−0.73*** (0.000)	−0.72*** (0.000)	−0.72*** (0.000)	−0.73*** (0.000)
TARGET	−0.07 (0.597)	−0.06 (0.690)	−0.06 (0.674)	−0.05 (0.712)
LN_GNI_CAP	0.02 (0.590)	0.02 (0.620)	0.02 (0.592)	0.02 (0.624)
GNI_CAP_GROWTH	−0.01 (0.106)	−0.01* (0.096)	−0.01 (0.113)	−0.01* (0.079)
WORLD_BANK_ GOVERNANCE	−0.14*** (0.007)	−0.14*** (0.005)	−0.14*** (0.005)	−0.14*** (0.005)
ENGLISH_LEGAL	−0.19 (0.261)	−0.18 (0.279)	−0.21 (0.225)	−0.21 (0.219)
BANK_REGULATION	0.42*** (0.000)	0.42*** (0.000)	0.41*** (0.000)	0.41*** (0.000)
TRUST	0.01*** (0.008)	0.01*** (0.004)	0.01*** (0.004)	0.01*** (0.006)
YEAR DUMMIES		YES		
Observations	4993	4993	4993	4993
Wald	615.90*** (0.000)	620.04*** (0.000)	620.80*** (0.000)	626.19*** (0.000)

*significant at 10% level, **significant at 5% level; ***significant at 1% level

Table 5: Determinants of MFI transparency: Investigating causality with structured equation modeling

This table etc. P values in parentheses. *significant at 10% level, **significant at 5% level; ***significant at 1% level. P values in parentheses. *significant at 10% level, **significant at 5% level; ***significant at 1% level. 4993 Observations

$$\text{TRANSPARENCY}_i = \alpha_i + \beta * \text{PORTFOLIO_QLTY} + \sum \beta_i * \text{MFI_PERFORMANCE}_i + \sum \beta_i * \text{MFI_CONTROLS}_i + \sum \beta_i * \text{GOVERNANCE}_i + \sum \beta_i * \text{MACROECONOMIC}_i + e_i$$

$$\text{PORTFOLIO_QLTY} = \alpha_i + \beta * \text{TRANSPARENCY} + \sum \beta_i * \text{MFI_PERFORMANCE}_i + \sum \beta_i * \text{MFI_CONTROLS}_i + \sum \beta_i * \text{MACROECONOMIC}_i + e_i$$

$$\text{TRUST} = \alpha_i + \beta * \text{WORLD_BANK_GOVERNANCE} + \beta * \text{ENGLISH_LEGAL} + e_i$$

	TRANSPARENCY	PORTFOLIO_QLTY	TRUST
TRANSPARENCY		0.10** (0.021)	
PORTFOLIO_QLTY	-0.06 (0.346)		
PROFITABILITY	-0.01 (0.627)	0.07*** (0.000)	
EFFICIENCY	-0.02 (0.256)	0.05*** (0.000)	
LEVERAGE	-0.00** (0.029)	0.000 (0.597)	
LNASSETS	0.23*** (0.000)	-0.02 (0.268)	
AGE	-0.09*** (0.003)	-0.04 (0.103)	
FOR_PROFIT	0.13*** (0.002)	-0.12*** (0.000)	
OUTREACH	0.01 (0.737)	0.03 (0.186)	
SUSTAINABILITY	0.01 (0.809)	0.42*** (0.000)	
REGULATED	0.09* (0.054)	0.05 (0.161)	
DEPOSITS	-0.38*** (0.000)	-0.01 (0.824)	
CHARTER_TYPE	-0.59*** (0.000)	0.08** (0.047)	
TARGET	-0.08* (0.061)	-0.09*** (0.008)	
LN_GNI_CAP	0.03*** (0.001)	0.00 (0.482)	
GNI_CAP_GROWTH	-0.01*** (0.000)	0.01*** (0.000)	
WORLD_BANK_GOVERNANCE	-0.07*** (0.000)		-3.00*** (0.000)
ENGLISH_LEGAL	-0.11** (0.026)		1.35** (0.012)
BANK_REGULATION	0.22*** (0.000)		
TRUST	0.01*** (0.000)		
INTERCEPT	-0.80*** (0.003)	-0.01 (0.981)	45.11*** (0.000)
LR Chi-square		1344.65** (0.023)	

*significant at 10% level, **significant at 5% level; ***significant at 1% level

Table 6: Determinants of MFI transparency: Robustness checks with propensity score matching

Table shows results propensity score matching with ordered logit regressions with standard errors clustered by microfinance institution, controlling for year fixed effects. Dependent variable, TRANSPARENCY, can take on discreet values of 0,1,2,3,4,5. Variance inflation factors for all variables in all models less than 10. Variables defined in Table 2. P values in parentheses. Unbalanced panel data for 2003–2015. 1336 MFIs from 73 countries (6370 MFI/year observations)

In Model 1, MFIs that accept deposits are a treatment group and MFIs that do not accept deposits are a control group. In Model 2, MFIs that are for-profit are a treatment group and MFIs that are nonprofit are a control group. We match each of these MFI pairs of treatment and control based on profitability, size and leverage using a non-replacement approach.

	TRANSPARENCY	
	1 DEPOSITS	2 PROFIT STATUS
PROFITABILITY	−0.00 (0.988)	0.02 (0.837)
PORTFOLIO_QLTY	0.16*** (0.001)	0.16*** (0.000)
EFFICIENCY	−0.07 (0.455)	−0.07 (0.468)
LEVERAGE	−0.00 (0.313)	−0.00 (0.896)
LNASSETS	0.32*** (0.000)	0.33*** (0.000)
AGE	−0.20** (0.041)	−0.20** (0.046)
FOR_PROFIT	0.28* (0.083)	0.29* (0.072)
OUTREACH	0.11 (0.360)	0.13 (0.297)
SUSTAINABILITY	−0.26** (0.046)	−0.28** (0.034)
REGULATED	0.07 (0.672)	0.05 (0.750)
DEPOSITS	−0.53*** (0.000)	−0.54*** (0.000)
CHARTER_TYPE	−0.74*** (0.000)	−0.71*** (0.000)
TARGET	−0.02 (0.907)	−0.01 (0.919)
LN_GNI_CAP	0.03 (0.452)	0.02 (0.559)
GNI_CAP_GROWTH	−0.01 (0.119)	−0.01 (0.149)
WORLD_BANK_ GOVERNANCE	−0.14*** (0.006)	−0.15*** (0.004)
ENGLISH_LEGAL	−0.21 (0.241)	−0.26 (0.132)
BANK_REGULATION	0.42*** (0.000)	0.41*** (0.000)
TRUST	0.01*** (0.006)	0.01** (0.011)
YEAR DUMMIES		
Observations	4713	4726
Wald	613.47*** (0.000)	597.73*** (0.000)

*significant at 10% level, **significant at 5% level; ***significant at 1% level

Appendix 1: Table of Pearson Correlation Coefficients: Independent Variables

Panel 1: MFI variables

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1 TRANSPARENCY	1														
2 PERFORMANCE	0.08	1.00													
3 PORTFOLIO_QLTY	0.07	0.55	1.00												
4 PROFITABILITY	0.05	0.80	0.22	1.00											
5 EFFICIENCY	-0.06	-0.66	-0.05	-0.27	1.00										
6 LEVERAGE	-0.03	-0.06	-0.01	-0.10	-0.01	1.00									
7 LNASSETS	0.25	0.19	0.01	0.20	-0.13	-0.02	1.00								
8 AGE	0.02	0.12	-0.01	0.16	-0.06	-0.05	0.25	1.00							
9 FOR_PROFIT	0.13	0.00	0.00	0.00	-0.01	-0.01	0.31	-0.17	1.00						
10 OUTREACH	0.19	0.20	0.03	0.10	-0.26	0.01	0.69	0.16	0.25	1.00					
11 SUSTAINABILITY	0.02	0.36	0.18	0.39	-0.13	-0.06	0.14	0.13	0.02	0.10	1.00				
12 REGULATED	0.02	0.11	0.03	0.13	-0.04	0.01	0.29	-0.07	0.36	0.18	0.08	1.00			
13 DEPOSITS	-0.18	0.03	0.00	0.04	-0.01	0.03	0.20	0.07	0.07	0.18	0.04	0.35	1.00		
14 CHARTER_TYPE	-0.15	0.09	0.01	0.14	0.00	0.01	0.20	0.06	0.10	-0.01	0.08	0.31	0.49	1.00	
15 TARGET	0.00	0.07	-0.01	0.20	0.10	0.00	0.16	0.10	0.02	-0.17	0.10	0.24	0.11	0.23	1.00

Panel 2: Country-level variables

	1	2	3	4	5	6
1 LN_GNI_CAP	1.00					
2 GNI_CAP_GROWTH	-0.15	1.00				
3 WORLD_BANK_GOVERNANCE	0.05	0.05	1.00			
4 ENGLISH_LEGAL	-0.05	0.27	-0.10	1.00		
5 BANK_REGULATION	-0.20	-0.06	0.02	-0.14	1.00	
6 TRUST	-0.32	-0.05	-0.30	0.06	0.27	1.00